# Muhammad Muizz Zafar

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**Data Engineer | ETL | Data Modeling | Data Quality | SQL Server 2022 | C# / ASP.NET Core**

Data engineer: ETL pipelines, data modeling, data profiling and data quality, SQL Server 2022 migration, and C#/ASP.NET Core development.

## Skills

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| Languages: | Python, SQL, GoLang, Java, C#, C++ |
| Data Engineering: | Data Warehousing, Data Modeling, Data Profiling, Data Quality, Data Governance, Apache Airflow, PyS |
| | AWS (S3, Lambda, Athena), MSSQL Server 2022, Oracle, Docker, Git, Linux, Datalake, ASP.NET Core |

## Education

**University of Ottawa** — Ottawa, ON
Bachelor of Applied Science, Software Engineering *GPA: 3.7/4.0* — Sept 2020 - Dec 2024

## Work experience

**Data Engineer** — May 2024 - Present
*Government Of Canada* — Ottawa, ON

- Designed logical and physical data models, defining schema and relationships for a centralized **Microsoft SQL Server** data warehouse supporting datasets of **20M+ records**.
- Built and maintained scalable **ETL pipelines** to extract data from **Oracle** databases, transform it using **SQL** and **Python**, and load it into semi-structured data in a JSON-based **data lake** layer and Microsoft SQL Server data warehouse, processing **5GB+** of data per pipeline run.
- Executed enterprise database migration by upgrading **SQL Server 2019 to SQL Server 2022** across **DEVTEST, UAT, and PROD** environments, migrating **50M+ records**, database schemas, stored procedures, triggers, and security configurations with zero downtime.
- Developed a reusable ETL validation framework using **Python** and **PySpark**, enabling automated validation of large datasets (**20M+ records**) across dev, uat, and prod environments, reducing data discrepancies by **60%**.
- Implemented data quality checks, scoring logic, and anomaly detection workflows to proactively identify inconsistencies, improving data reliability and reducing production data issues by **50%**.
- Orchestrated and scheduled data workflows using **Apache Airflow**, automating pipeline execution and dependency management, reducing manual intervention by **70%** and improving pipeline reliability.
- Supported data migration projects, validating data quality and integrity during migration and improving consistency across systems.
- Developed and optimized complex **SQL queries and transformations**, reducing processing time for large-scale ETL jobs by **35 minutes**.
- Built **Power BI and Streamlit dashboards** for pipeline performance and data quality, helping stakeholders identify issues **2x faster**.

**Full Stack Developer Intern** — September 2023 - April 2024
*University of Ottawa* — Ottawa, ON

- Redesigned the University of Ottawa's emergency alert website with **C#** and **Angular**, introducing a modern, interactive UI. Used by over **20,000** students daily.
- Optimized existing **C#** REST APIs by reducing **SQL** query execution time for event data lookup by **30 %** and lines of code by **40 %**
- Revamped the event registration and admin webpages by using **Angular** material UI components to improve UI layout and reducing redundant lines of code by almost **50 %**

**Data Engineer Intern** — May 2023 - August 2023
*Trend Micro Inc.* — Ottawa, ON

- Designed and implemented a serverless data pipeline using **AWS Lambda, S3, and Athena**, leveraging **AWS SAM** and **CloudFormation** for infrastructure-as-code and automated deployment.
- Enabled scalable data querying workflows by integrating **Amazon Athena** to query data stored in S3, reducing manual data processing efforts.
- Configured **AWS CloudWatch** and **Splunk** monitoring to track pipeline performance, logs, and failures, enhancing observability and system reliability.

**Cloud Developer Intern** — September 2022 - December 2022
*Trend Micro Inc.* — Ottawa, ON

- Developed serverless data processing pipelines using **AWS Lambda** to analyze client files, images, and code stored in **Amazon S3**, performing vulnerability scans using Anchore APIs and Grype.
- Implemented concurrent processing and robust error handling in **GoLang** using goroutines within Lambda functions, improving reliability and reducing data corruption in high-throughput environments.
- Enabled scalable data querying workflows by integrating **Amazon Athena** to query data stored in S3, reducing manual data processing efforts.

**Full Stack Developer Intern** — January 2022 - April 2022
*Applied Mind Inc.* — Ottawa, ON

- Containerized a full-stack application with **Docker** for reliable **production deployments**, and automated event detection by adding events to **PostgreSQL** using **Python** and **GraphQl**, reducing testing time by **10 seconds**.